Quiz 2: Practice

15 minutes 10 points

Name: _____ **UW NetID:** _____ **Campus ID:** _____

Final answers **For TA:** Total _____ / 10

1(a)	1(b)	1(c)	2(a)	2(b)

Lectures 4–7. Not graded. Closed book, with the formulas below. Work individually; no personal notes or AI tools. **Write final answers in the table above:** numbers for 1(a)–(c), one letter (A–D) for each of 2(a)–(b). Exact fractions or decimals are accepted. Each part is worth 2 points; no partial credit.

Reference. For a value table V and a deterministic policy π:

$$Q_V(s,a) = r(s,a) + \gamma \sum_{s'} P(s' \mid s,a) V(s'),$$

$$(\mathcal{T}V)(s) = \max_a Q_V(s,a), \qquad V^\pi(s) = Q_{V^\pi}\big(s, \pi(s)\big).$$

Exact policy iteration: $E_{k+1} \le \gamma E_k$, where $E_k = \|V^\star - V^{\pi_k}\|_\infty$ and $\|x\|_\infty = \max_s |x(s)|$.

1. Value iteration, evaluation, and convergence — 6 points

For **(a)–(b)**, use this continuing MDP with $\gamma = \frac{1}{2}$. Each step pays the listed reward, then moves. All vectors and policy pairs are in state order **(X,Y)**. The parts use separate value tables or policies.

State s	Action	$r(s,a)$	$P(\mathrm{X} \mid s,a)$	$P(\mathrm{Y} \mid s,a)$
X	a	1	1	0
X	b	-4	0	1
Y	a	10	$1/2$	$1/2$
Y	b	-1	1	0

(a) Value iteration. Given $V_k = (4, 8)$, compute $V_{k+1}(\mathrm{Y})$ for $V_{k+1} = \mathcal{T}V_k$. Use the supplied old values.

(b) Policy evaluation. A PI run has current policy $\pi = (b, b)$. Compute its *exact* infinite-horizon value $V^\pi(\mathrm{Y})$.

(c) Convergence. In an independent exact PI run, $\gamma = \frac{1}{2}$ and $E_0 \le 12$. What numerical upper bound on E_2 does the reference rate give after two greedy improvements, with exact policy evaluations?

2. Policy improvement and stopping — 4 points

(a) The table gives exact action values for a policy π_0. Which policy $(\pi_1(\mathrm{X}), \pi_1(\mathrm{Y}))$ does one greedy improvement produce?

State s	$Q^{\pi_0}(s,a)$	$Q^{\pi_0}(s,b)$
X	2	3
Y	14	0

A. (a,a) **B.** (a,b)
C. (b,a) **D.** (b,b)

(b) Consider two stopping checks in finite discounted MDPs. PI keeps the current action whenever it is a maximizer.

I. Exact PI: after exact evaluation, $\mathcal{T}V^\pi = V^\pi$; return π.
II. VI: $V_{k+1} = V_k$; return a policy greedy with respect to this value table.

Which checks guarantee that the returned policy is optimal?

A. I only **B.** Neither **C.** II only **D.** Both I and II